

Annan Yu

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POSITION

AMAZON.COM INC.

NEW YORK, NY

Applied Scientist, June 2026-present

EDUCATION

CORNELL UNIVERSITY

Ithaca, NY

M.S. Applied Mathematics, 2024

Ph.D. Applied Mathematics, 2026

- **Advisor:** Prof. Alex Townsend
- **Dissertation Title:** Implicit Regularization and Inductive Biases in Neural Network Training

VANDERBILT UNIVERSITY

Nashville, TN

B.S. Computer Science and Mathematics, 2021

Summa Cum Laude

Highest Honors in Mathematics, Honors in Computer Science

- **GPA:** 4.00/4.00
- **Senior Thesis Title:** Geometric Analysis of Quasilinear Wave Equations

RESEARCH INTERESTS

Deep Learning, Sequence Modeling, Time Series Analysis, Numerical Analysis, Reduced-order Modeling

RESEARCH EXPERIENCE

AMAZON AWS DEEP LEARNING LAB, Santa Clara, CA, USA

Applied Scientist Intern, May 2025 - August 2025 (full-time), August 2025 - October 2025 (part-time)

- Worked as an applied scientist to investigate time-series forecasting models.
- Designed more efficient models and investigated the effect of each design choices in time-series foundation models.
- Published two papers in ICLR 2026 as first author.

LAWRENCE BERKELEY NATIONAL LABORATORY, Berkeley, CA, USA*Student Assistant, May 2023 - August 2023, May 2024 - August 2024*

- Supervised by Professor Michael Mahoney and Dr. Benjamin Erichson.
- Analyzed different initialization schemes of state-space models and improved them.
- Published three papers in ICLR 2024 and 2025 as first author.

THE FIELDS INSTITUTE FOR RESEARCH IN MATHEMATICAL SCIENCES, Toronto, ON, Canada*Summer Research Internship, July 2020 - August 2020*

- Supervised by Professor Angèle Foley.
- Defined the new concept of H-chromatic symmetric function, conducted numerical experiments, and formulated and proved its properties.
- Published the paper *On H-chromatic symmetric functions*.

MATH DEPARTMENT AT VANDERBILT UNIVERSITY, Nashville, TN, USA*Summer Research Internship, May 2019 - Jul 2019, May 2020 - Present*

- Assistant to Professor Larry Schumaker.
- Implemented over 10000 lines of MATLAB code.
- Designed and analyzed algorithms in a forthcoming book on spline functions.
- Published the paper *Polynomial Splines on Curved Triangulations*.

PUBLICATIONS AND PREPRINTS

[1] Annan Yu, Danielle C. Maddix, Boran Han, Xiyuan Zhang, Abdul Fatir Ansari, Oleksandr Shchur, Christos Faloutsos, Andrew Gordon Wilson, Michael W. Mahoney, Yuyang Wang, *Understanding Transformers for Time Series: Rank Structure, Flow-of-ranks, and Compressibility*, ICLR, 2026.

[2] Annan Yu, Danielle C. Maddix, Boran Han, Xiyuan Zhang, Abdul Fatir Ansari, Oleksandr Shchur, Christos Faloutsos, Andrew Gordon Wilson, Michael W. Mahoney, Yuyang Wang, *Understanding the Implicit Biases of Design Choices for Time Series Foundation Models*, ICLR, 2026.

[3] Anil Damle, Silke Glas, Alex Townsend, Annan Yu, *How to reveal the rank of a matrix?*, Linear Algebra and its Applications, vol 731, 2026.

[4] Yihan Wang, Lujun Zhang, Annan Yu, N. Benjamin Erichson, Tiantian Yang, *A deep state-space model for rainfall-runoff simulations*, Water Resources Research, vol 61, 2025.

[5] Annan Yu, N. Benjamin Erichson, *Block-biased mamba for long-range sequence processing*, NeurIPS, 2025.

- [6] Annan Yu, Dongwei Lyu, Soon Hoe Lim, Michael W. Mahoney, N. Benjamin Erichson, *Tuning frequency bias of state space models*, ICLR (spotlight), 2025.
- [7] Soon Hoe Lim, Yijin Wang, Annan Yu, Emma Hart, Michael W. Mahoney, Xiaoye S. Li, N. Benjamin Erichson, *Elucidating the design choice of probability paths in flow matching for forecasting*, Transactions on Machine Learning Research, 2025.
- [8] Annan Yu, Michael W. Mahoney, N. Benjamin Erichson, *HOPE for a robust parameterization of long-memory state space models*, ICLR, 2025.
- [9] Annan Yu, Arnur Nigmatov, Dmitriy Morozov, Michael W. Mahoney, N. Benjamin Erichson, *Robustifying state-space models for long sequences via approximate diagonalization*, ICLR (spotlight), 2024.
- [10] Annan Yu, Alex Townsend, *Leveraging the Hankel norm approximation and block-AAA algorithms in reduced order modeling*, Numerical Linear Algebra with Applications; e2555, 2024.
- [11] Annan Yu, Alex Townsend, *On the stability of unevenly spaced samples for interpolation and quadrature*, BIT Numerical Mathematics, vol 63(23), 2023.
- [12] Annan Yu, Yunan Yang, Alex Townsend, *Tuning frequency bias in neural network training with nonuniform data*, ICLR, 2023.
- [13] Nancy Mae Eagles, Angèle M. Foley, Alice Huang, Elene Karangozishvili, Annan Yu, *On H-chromatic symmetric functions*, The Electronic Journal of Combinatorics, vol 29(1), 2022.
- [14] Larry Schumaker, Annan Yu, *Approximation by polynomial splines on curved triangulations*, Computer Aided Geometric Design, vol 92, 2022.
- [P1] Annan Yu, Dongwei Lyu, N. Benjamin Erichson, *Continuity Laws for Sequential Models*, arXiv: 2605.08539.
- [P2] Aditi Gupta, Soon Hoe Lim, Annan Yu, N. Benjamin Erichson, *Sharpen Your Flow: Sharpness-Aware Sampling for Flow Matching*, arXiv: 2605.11547.
- [P3] Yihan Wang, Annan Yu, Lujun Zhang, Charuleka Varadharajan, N. Benjamin Erichson, *HydroDiffusion: Diffusion-based probabilistic streamflow forecasting with a state space backbone*, arXiv:2512.12183.
- [P4] Annan Yu, Chloe Becquey, Diana Halikias, Matthew E. Mallory, Alex Townsend, *Arbitrary-depth universal approximation theorems for operator neural networks*, arXiv:2109.11354.

REVIEWER EXPERIENCE

ICLR 2024: 4 papers

ICLR 2025: 3 papers

ICLR 2026: 5 papers

ICML 2023: 3 papers

ICML 2024: 7 papers

ICML 2025: 6 papers

ICML 2026: 2 papers

NeurIPS 2023: 1 paper

NeurIPS 2024: 6 papers (Best Reviewer Award)

NeurIPS 2025: 3 papers

NeurIPS 2026: 4 papers

TMLR: 5 papers

INVITED TALKS

Understanding Transformers for time series: Rank structure, flow-of-ranks, and compressibility, International Conference on Learning Representations, Brazil, 2026. (poster)

Understanding the implicit biases of design choices for time series foundation models, International Conference on Learning Representations, Brazil, 2026. (poster)

Block-biased Mamba for long-memory processing, Neural Information Processing Systems, San Diego, USA, 2025. (poster)

Why should machine learning researchers care about low-rank structures?, SIAM NY-NJ-PA Section Conference, Penn State University, State College, USA, 2025.

Understanding Transformers for time series, Mathematical Sciences Colloquium, Rensselaer Polytechnic Institute, Troy, USA, 2025. (1-hour colloquium invited by Andrew Horning)

Remember me: a fine-grained analysis of input-selective Mamba models for temporal data, SIAM Conference on Applications of Dynamical Systems (DS25), Denver, USA, 2025.

How to train a rational function?, BIRS Workshop: Challenges, Opportunities, and New Horizons in Rational Approximation, Banff, USA, 2025.

State space models for forecasting time series, SIAM Conference on Computational Science and Engineering (CSE25), Fort Worth, USA, 2025.

Deep learning with state space models, SIAM Student Chapter Seminar, Virginia Tech, USA, 2025. (online)

Training an LTI system without an objective: a numerical analyst's perspectives on state-space models, Lunch Talk invited by Dr. Mert Pilanci, Stanford University, Stanford, USA, 2025.

Training an LTI system without an objective: a numerical analyst's perspectives on state-space models, Mid-Atlantic Numerical Analysis Day, Temple University, Philadelphia, USA, 2024.

How does a machine learn sequences: an applied mathematician's guide to transformers, state-space models, mamba, and beyond, SIAM Conference on Mathematics of Data Science (MDS24), Atlanta, USA, 2024. (2-hour tutorial)

Learning without an objective: a mathematical guide to state-space models, Amazon AWS AutoML/Forecasting Reading Group Meeting, 2024. (online)

Representations and evolution of linear time-invariant systems in state-space models, 26th International Symposium on Mathematical Theory of Networks and Systems, University of Cambridge, UK, 2024.

Rectifying unstable rational representations in sequence models, SIAM Conference on Applied Linear Algebra, Sorbonne Université, Paris, France, 2024.

Robustifying state-space models for long sequences via approximate diagonalization, International Conference on Learning Representations, Vienna, Austria, 2024.

How to avoid HiPPOs?, SCAN Seminar, Cornell University, USA, 2024

Robustifying state-space models via approximate diagonalization, SciDAC AI monthly synchronization, Lawrence Berkeley National Laboratory, USA, 2024. (online)

Linear time-invariant systems in machine learning (50-minute tutorial), SIAM New York-New Jersey-Pennsylvania (NNP) Section Annual Meeting, New Jersey Institute of Technology, USA, 2023.

Leveraging the Hankel norm approximation and data-driven algorithms in reduced order modeling, Numerical Analysis in the 21st Century, University of Oxford, UK, 2023.

Leveraging the Hankel norm approximation and block-AAA algorithms in reduced order modeling, SIAM Southeastern Atlantic (SEAS) Section Annual Meeting, Virginia Tech, USA, 2023.

Tuning frequency bias in neural network training, International Conference on Learning Representations, 2023. (online)

Tuning frequency bias in neural network training, Workshop on Numerics and Acoustics, Imperial College London, UK, 2022.

H-chromatic symmetric functions, 2020 FUSRP Mini-Conference, Fields Institute for Research in Mathematical Sciences, Canada, 2020. (online)

TEACHING EXPERIENCE

MATH DEPARTMENT AT CORNELL UNIVERSITY, Ithaca, NY

Recitation Teaching Assistant, August 2022 - Present

- Fall 2023: MATH 6110, Real Analysis (1 recitation section)
- Fall 2022: MATH 2310, Linear Algebra with Applications (2 recitation sections)

MATH DEPARTMENT AT VANDERBILT UNIVERSITY, Nashville, TN

Grader/Teaching Assistant, Jan 2018 - Dec 2020

- Fall 2020: MATH 3620, Introduction to Numerical Mathematics
- Spring 2020: MATH 3100, Introduction to Analysis
- Spring 2020: MATH 3620, Introduction to Numerical Mathematics
- Fall 2019: MATH 3200, Introduction to Topology
- Spring 2018: MATH 2300, Multivariable Calculus

EECS DEPARTMENT AT VANDERBILT UNIVERSITY, Nashville, TN

Grader/Teaching Assistant, Aug 2018 - Dec 2019

- Fall 2019: CS 3251, Intermediate Software Design
- Fall 2018: CS 1101, Programming and Problem Solving

HONORS & AWARDS

RICHARD J. LARSEN AWARD FOR ACHIEVEMENT IN UNDERGRADUATE MATHEMATICS

- Conferred by the Department of Mathematics at Vanderbilt University in April 2021.
- The award, along with a check for \$500, is presented each spring to the senior math major judged by the faculty to have excelled in all aspects of undergraduate mathematics.

WILSON L. AND NELLIE PYLE MISER AWARD

- Conferred by the School of Engineering at Vanderbilt University in April 2021.
- This award is given to the senior who has excelled in all aspects of mathematics during the student's undergraduate career.